

Growth of rutile GeO₂ nanoneedles supported by in situ microscopy

Alexey S.T. Rybakov^{*}, Vu Anton Lie, Louis Mirecki, Jörg August Becker

Institute of Physical Chemistry and Electrochemistry, Leibniz University Hannover, Hannover, Germany

ARTICLE INFO

Communicated by Pierre Muller

ABSTRACT

Rutile GeO₂ nanoneedles were grown using a method involving eutectics of the Ge-GeO₂ system and chemical transport reactions. The synthesis was carried out at 1095–1215 K in a miniature cell, which is evacuated and sealed, enclosing all products that are generated in reactions. In this cell, one can observe the processes throughout all stages of the method in situ via video microscopy. Differences in the process development depending on the heating time and mass ratio of reagents were revealed. Depending on the latter condition, two ways for growing nanoneedles have been proposed. The needles and other reaction products were studied ex situ using SEM, dark-field and bright-field TEM, HRTEM, SAED, EDX and Raman spectroscopy. The nanoneedles are single crystals without any amorphous surface layer. They are rutile GeO₂ and belong to the tetragonal crystal system. Their longitudinal growth direction is [001]. The spacings between lattice planes in the longitudinal and lateral directions are found to be 2.8 Å and 3.2 Å respectively. At higher temperatures, larger micrometre-sized crystals of rutile GeO₂ with various morphologies are formed.

1. Introduction

Nanomaterials differ greatly in their properties from those of the bulk due to their small sizes. Size-dependent properties are optical, electrical and magnetic properties, melting point, ionisation potential, hardness, catalytic activity and selectivity, etc. [1] A wide range of methods makes it possible to grow nanostructures of various geometries. In recent decades, numerous studies have focused on one-dimensional germanium dioxide (GeO₂) structures, such as nanowires, nanotubes, nanoneedles, nanorods, etc. For their growth, methods have been proposed which may include the following processes: carbothermal reduction [2], thermal evaporation [3–5], laser ablation [6], thermal oxidation [2,7,8], non-catalysed [3] and self-catalysed [4] or metal-catalysed [5] growth of nanostructures, local anodic oxidation [9], electrodeposition [10], liquid phase deposition [11], etc. Most methods produce hexagonal GeO₂ (GeO_{2(hex.)}), a high-temperature polymorph of germanium dioxide, rather than tetragonal GeO₂ (GeO_{2(tet.)}), a high-pressure polymorph. However, the latter is thermodynamically stable at ambient conditions, while the former is metastable [12]. They are both predominant polymorphic modifications of crystalline GeO₂. Tetragonal GeO₂ is the rutile-type GeO₂ (space group P4₂/mnm, No. 136) with Ge ions having six-fold coordination. The crystal structural type of hexagonal GeO₂ is α-quartz (enantiomorphic space-group pair P3₁21, No. 152, for right-handed and P3₂21, No. 154, for left-handed

α-quartz [13,14]). Ge ions in it are four-fold coordinated [12,15–17].

Tetragonal GeO₂, which is stable at ambient temperature and pressure, undergoes a phase transition GeO_{2(tet.)} → GeO_{2(hex.)} at approx. 1300 K under room pressure [16,18,19]. According to Deringer et al. [12] the increased stability of GeO_{2(hex.)} at high temperatures and ambient pressure arises from the higher entropy at rising temperature, which leads to a more negative Gibbs free energy than in the case of GeO_{2(tet.)}. However, the reverse phase transition GeO_{2(hex.)} → GeO_{2(tet.)} on cooling is hardly possible without a catalyst [20]. If hexagonal GeO₂ is heated further, it starts melting at 1389 K [21]. As the melt is cooled down, however, a glass network is formed. To obtain the transition from glassy to crystalline phase, alkali oxides, fluorides or carbonates can be used as a glass network breaker [22].

Although GeO_{2(hex.)} is metastable at lower temperatures, as demonstrated by Laubengayer et al. [16], in the pure, dry state its transition into stable GeO_{2(tet.)} is quite sluggish. They propose the use of water or flux as a catalyst to accomplish the transition at lower temperatures. In a neutron diffraction study of the thermal stability of GeO_{2(hex.)}, Haines et al. [23] confirmed that its phase transition only begins to be significant above approx. 1000 K. They suggest that the phase transition is influenced not only by the sample environment but also by the size of GeO_{2(hex.)} crystals. While slow kinetics even at high temperatures is probably the dominant factor for the bulk material, in the case of nanoscale crystals with much greater surface-area-to-volume ratios,

^{*} Corresponding author at: Institute of Physical Chemistry and Electrochemistry, Leibniz University Hannover, Callinstr. 3A, 30167 Hannover, Germany.
E-mail address: alexey.rybakov@pci.uni-hannover.de (A.S.T. Rybakov).

surface effects should also be considered. These effects may include surface defects such as OH groups as well as highly increased Laplace pressure. Numerous studies, such as [15,19] and references therein, show that at high pressures of the order of GPa, the transition from $\text{GeO}_2(\text{hex.})$ to $\text{GeO}_2(\text{tet.})$ occurs, which is however extremely sluggish even in combination with heating to high temperatures.

The realisation of the transition from hexagonal to tetragonal phase of GeO_2 is challenging. In recent years, methods for growth of the latter have been developed as well. Tetragonal GeO_2 has attracted increased interest due to its direct [24] and ultrawide band gap of approx. 4.7 eV, and its stability at high temperatures and high pressures. Therefore, it has potential for high-power devices and is a promising candidate for applications in semiconductors, optoelectronics, photocatalysis and gas sensing [25]. For these purposes, $\text{GeO}_2(\text{tet.})$ can be grown as a thin film on various substrates using deposition methods, e.g., MOCVD, MBE, CVD, and sputtering, which are compared with each other and summarised in a review [25]. References to original studies are provided therein. Also, it can be grown as centimetre-sized single crystals synthesised by a flux method [22,26]. Jing et al. [11] proposed a method for growing $\text{GeO}_2(\text{hex.})$ particles and films, as well as $\text{GeO}_2(\text{tet.})$ nanorods in liquid phase deposition process. Wang et al. [27] synthesised micrometre-sized tetragonal GeO_2 crystals using a hydrothermal method and demonstrated its capability for photocatalytic water splitting. Recently, a map of morphological transformation routes of tetragonal GeO_2 nanoparticles was elaborated by Laranjeira et al. [28]. They could also provide the transformation routes of the crystals from [11] and [28] starting from the $\text{GeO}_2(\text{tet.})$ crystal shape in thermodynamic equilibrium. Shinde et al. [8] presented a method for growing tetragonal GeO_2 structures with flower-like morphology composed of nanometre- and micrometre-sized rods using thermal oxidation. Chemical vapour transport reactions can also be involved to grow pure single crystals of different morphologies, as summarised in the comprehensive handbook by Binnewies et al. [29] on this type of chemical reactions. For example, GeO_2 can be transported with the help of chlorine to produce micrometre-sized $\text{GeO}_2(\text{tet.})$ needle-like crystals at temperatures of approx. 1080 K [30]. However, this not only requires seeds of rutile GeO_2 but also a mineraliser.

In this study, we present methods for growing tetragonal GeO_2 nanoneedles under well-controlled thermochemical conditions in a closed reaction cell with a volume below ten cubic millimetres, containing cubic Ge and hexagonal GeO_2 as reagents, with masses up to a few tens of micrograms and of several milligrams respectively. The novelty relative to previous works [8,11,30], in which rod-like structures were also grown, is that our methods of growing $\text{GeO}_2(\text{tet.})$ nanoneedles utilise only Ge and $\text{GeO}_2(\text{hex.})$, and a closed, pre-evacuated cell ensures that all reaction products remain inside it. As discussed above, direct transformation from hexagonal to tetragonal phase of GeO_2 is challenging, but our research suggests a way to overcome this barrier by taking advantage of eutectics of the Ge- GeO_2 system. With optimally adjusted parameters, it becomes possible to grow large quantities of $\text{GeO}_2(\text{tet.})$ nanoneedles. When the parameters are varied, tetragonal GeO_2 crystals with other morphologies can be also formed. This study also further reveals the potential of the Ge- GeO_2 system, since in the same one experiment, i.e., in the same cell, besides germanium/germanium oxide microneedles, which were the focus of our previous study [31], $\text{GeO}_2(\text{tet.})$ nanoneedles can also be grown. Furthermore, via in situ video microscopy, the processes in the cell throughout all stages of the methods are observable. We also succeeded in studying the reaction products ex situ with field emission scanning electron microscopy (FE-SEM), energy-dispersive X-ray (EDX) and Raman spectroscopy, field emission transmission electron microscopy (FE-TEM), and selected area electron diffraction (SAED). These experimental techniques were complemented with simulations to determine the morphology, crystal structure and the longitudinal growth direction of the nanoneedles.

2. Experimental setup

The microreactor for growing tetragonal GeO_2 nanoneedles is similar to reactor already described in [32] for growing SiO_2 nanowhiskers via a vapour-liquid-solid mechanism. The reactor was further improved for our previous research on composite germanium/germanium oxide microneedles [31] and described in more detail therein. In the present study, the focus is on tetragonal GeO_2 nanoneedles, so the reactor was again slightly modified to set the necessary conditions for their growth and to systematically study this process. The changes relative to [31] are in chemical transport cell design, mass and mass ratio of the reagents, and heating and cooling procedures. The following chapter discusses this.

2.1. Apparatus

The central part of the microreactor is shown in Fig. 1 a. During the experiments, it is operated in a high vacuum chamber at a pressure of 10^{-7} – 10^{-6} mbar, which is reached by means of a turbomolecular pump (Edwards R72202003) backed by a scroll pump (Varian SH-100). The chemical transport cell is built into two heating units, the upper and the lower boron nitride (BN) heater. For this purpose, two BORALECTRIC® heaters (Momentive Performance Materials Inc.) are used in the setup. Each of them is driven by a separate DC power supply unit (FUG NTN 1400–65) controlled with a LabVIEW (National Instruments) software. By controlling the applied voltage for each heater, it is possible to provide a stable and accurate temperature difference in the cell.

The chemical transport cell itself is composed of three glass plates which are stacked on top of each other. The middle plate acts as a spacer and is made of polished Corning® EAGLE XG® (alkaline earth borosilicate) glass (MTI Corporation) with a softening point ($10^{7.6}$ poises) of 1244 K [33]. It is 0.7 mm thick and has planar dimensions of approx. 9 mm × 9 mm. There is a hole with a diameter of approx. 4 mm in the centre of the spacer. Two 0.5 mm-thick polished fused silica (FS) plates (MTI Corporation) with a size of 10 mm × 10 mm are used to cover the spacer from above and below to create a small reaction chamber with a volume of 9 mm³. This chamber can be filled with reagents – Ge and GeO_2 . The cell becomes sealed once it is heated to a temperature of 973–1073 K.

The chemical transport cell is sandwiched between two boron nitride (BN) fixing plates with dimensions of approx. 10 mm × 10 mm × 1 mm and a small, drilled hole in the centre of one narrow side, so that a thermocouple (TC) of type C can be inserted in each plate. Fixing plates hold both TC in their proper position. In the following study, they are designated as BN TC plates. The LabVIEW software is used to measure the temperature of the FS plates as a function of time by means of TC. BN frame, BN cover and BN bottom fix the entire setup in place. The BN cover and bottom, both heaters and both BN TC plates have a central opening of approx. 2 mm for illumination of the cell from above and below. The processes in the cell can be observed in situ through the upper opening via a video microscope (microscope Olympus SZX16 with camera AVT Mako G-192C).

2.2. Preparation of chemical transport cells

The cell is composed of two FS plates and one Corning® EAGLE XG® spacer. The lower FS plate has a circular notch with an average diameter of 3.5 mm, a width of 0.6 mm and a depth of 0.25 mm around its centre. The notch makes it easier to open the cell after the experiment. It can also be used for the spatial separation of reagents (Fig. 1 c) as described in more detail below.

The reagents used in our study are hexagonal GeO_2 powder ($\geq 99.99\%$, Sigma-Aldrich) and Ge splinters. The latter are produced by breaking monocrystalline Ge substrates ($> 99.9953\%$, undoped, MTI Corporation) into small pieces. Since the powder is hygroscopic, it is permanently stored in a drying cabinet at 473 K, from which it is only taken out

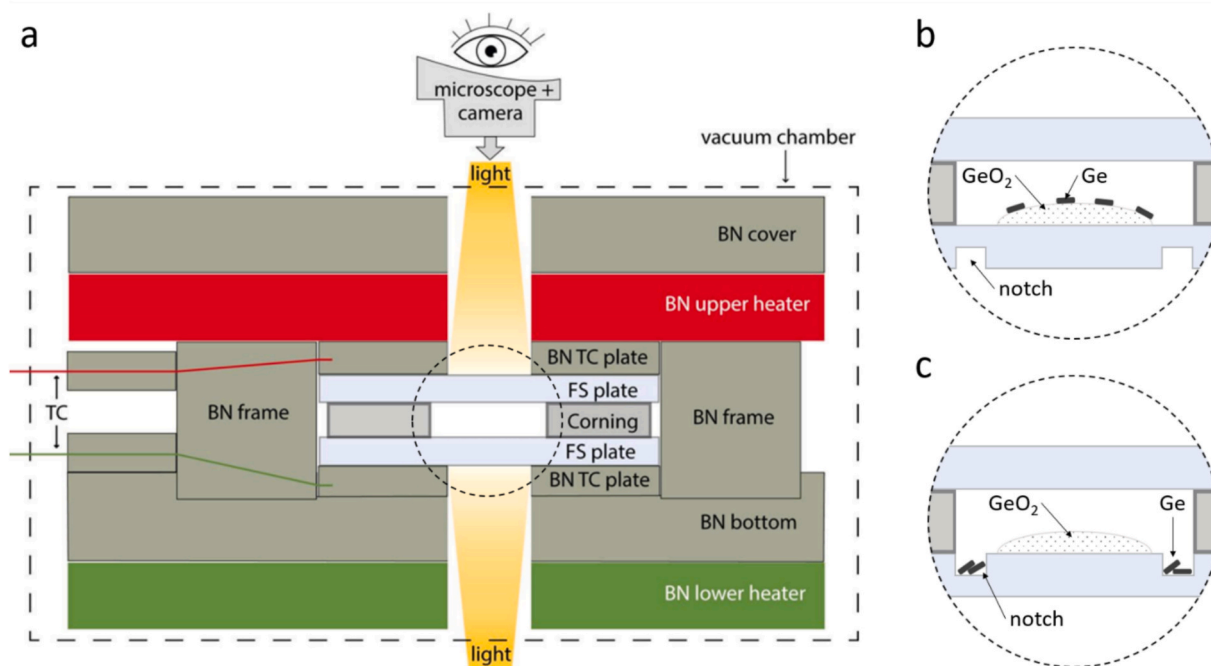


Fig. 1. **a** Vertical cross-section of the central part of the microreactor for chemical transport reactions. All details regarding the device components and cell preparation are provided in the text. BN stands for boron nitride, FS for fused silica and TC for thermocouple. Two different reagent arrangements were used in experiments in the reaction chamber (circled by the dashed line): **b** contact case (all tiny Ge splinters on a small pile of the hexagonal GeO_2 powder) and **c** non-contact case (Ge splinters in the notch of the lower FS plate).

just before the experiment. Two mass ratios of Ge to $\text{GeO}_{2(\text{hex.})}$, $m_{\text{Ge}} : m_{\text{GeO}_2}$, of approx. 1:4 and 1:12 are used. For both cases, the powder mass is set to 1100–1300 μg . First, two to five Ge splinters, each with a mass of 10–70 μg , are weighted with a total mass required for a certain mass ratio. Second, the powder is placed in the centre of the lower FS plate and distributed with a spatula in a layer with a width of about 3 mm. There are two possible arrangements of the reagents in the cell: contact case (Fig. 1 b) and non-contact case (Fig. 1 c). In the first case, all Ge splinters are on the GeO_2 powder, and the notch is outside the reaction chamber, while in the second case, the notch is inside the chamber and all Ge splinters are placed in it. In the cell with only Ge or only GeO_2 , no crystal growth could be observed, indicating the need for both reagents to grow the crystals.

Next, the chemical transport cell is fully assembled, fixed in the microreactor and placed in a vacuum chamber where the remaining gases are evacuated out of the cell. The detailed description of all these steps is given in our previous study [31].

The choice of these two mass ratios is based on the following considerations. The eutectic of the Ge- $\text{GeO}_{2(\text{hex.})}$ system is characterised by $T_{\text{EP, hex.}} = 1127 \text{ K}$ and a GeO_2 mole fraction of $x_{\text{GeO}_2}^{\text{EP, hex.}} = 0.765$ [18]. The presence of a liquid Ge- GeO_2 alloy is a necessary condition for the formation of GeO_2 nanoneedles. To study how much alloy is required, two mole fractions are selected: $x_{\text{GeO}_2} \approx x_{\text{GeO}_2}^{\text{EP, hex.}}$ and $x_{\text{GeO}_2} > x_{\text{GeO}_2}^{\text{EP, hex.}}$. In the first case, all or almost all of the GeO_2 powder is involved in the formation of the alloy. The appropriate mass ratio is 1:4, which corresponds to $x_{\text{GeO}_2} = 0.735$. In the second case, there is an excess of the powder, and only part of it can be used to form the alloy. The mass ratio with a large excess is 1:12, which corresponds to $x_{\text{GeO}_2} = 0.893$. In addition to mass ratio, the amount of alloy is also affected by heating and cooling procedures, which is discussed in the following chapter.

2.3. Heating and cooling procedures

Before heating, the cell remains unsealed, despite its polished surfaces. Above 973–1073 K, the soft spacer is adhered to both FS plates,

causing sealing of the cell and enclosing all products generated in reactions. Only the upper heater is used for heating the cell, while the power supply for the lower heater is completely turned off. This provides a temperature of the upper FS plate, $T_{\text{upper TC}}$, which is always higher than that of the lower plate, $T_{\text{lower TC}}$. The eutectic point of the Ge- GeO_2 (hex.) system, $T_{\text{EP, hex.}}$, is at 1127 K [18]. The melting point of Ge, $T_{\text{MP Ge}}$, equals 1211 K [21]. The maximum temperature on the lower and the upper FS plates is set to be 7–15 K below $T_{\text{EP, hex.}}$ and $T_{\text{MP Ge}}$, respectively. The exemplary temperature profile, i.e., $T_{\text{upper TC}}$ and $T_{\text{lower TC}}$ during the experiment, is represented in Fig. 2. Throughout the entire heating and cooling procedures, the changes in solid and liquid phases in the reaction chamber were observed in situ and recorded by means of a video microscope.

The entire procedure can be divided into three stages, which are discussed in detail below. In addition to a general overview of the stages, they are also considered using an example of two cells with $m_{\text{Ge}} : m_{\text{GeO}_2}$ of 1:4 and 1:12, shown in Fig. 3 a–d and e–h respectively. In both cells, the reagents are in contact case. The green arrow points to the same area on the GeO_2 powder so that one can trace the processes involved during all stages. The red arrow indicates Ge splinter or the position where it was before its complete sublimation or vaporisation.

The errors of $T_{\text{upper TC}}$ and $T_{\text{lower TC}}$ are $\pm 8 \text{ K}$ and $\pm 7 \text{ K}$, respectively. In addition to these uncertainties, the temperature difference between both FS plates, ΔT , can vary within a range. This variation is primarily due to changes in the contact surfaces between individual parts of the microreactor (Fig. 1 a) each time a new cell is installed. As can be concluded from average temperature values in Table 1, ΔT can range from a minimum of 75 K to a maximum of 100 K. According to the uncertainty analysis, the temperature difference during the second stage is equal to $90 \pm 8 \text{ K}$. Since the reagents lie on the lower FS plate, in the following discussions, as a default, $T_{\text{lower TC}}$ is used as a temperature throughout. To estimate $T_{\text{upper TC}}$ during the second stage, one can use the value of ΔT specified above. Only where necessary, both $T_{\text{upper TC}}$ and $T_{\text{lower TC}}$ are given.

Stage 1: heating of the cell

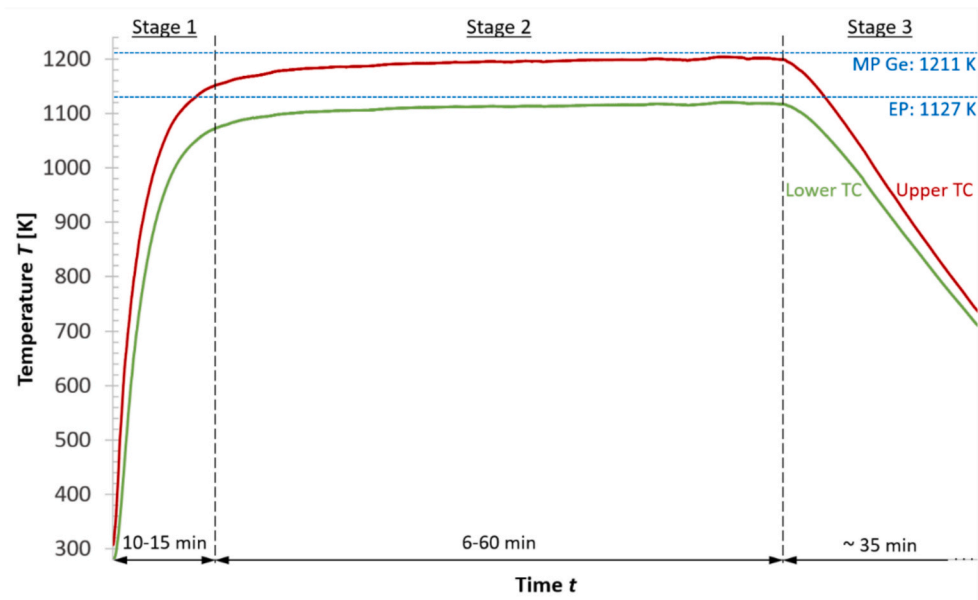


Fig. 2. Example of the temperature profile, i.e., temperature as a function of time, on the lower and upper fused silica (FS) plates during cooling and heating procedures. The profile can be divided into three stages: stage 1 (heating of the chemical transport cell to 1073 K on the lower FS plate), stage 2 (slow formation of a liquid alloy on the GeO_2 powder), stage 3 (cooling of the cell). The melting point (MP) of Ge of 1211 K [21] and the eutectic point (EP) of Ge- $\text{GeO}_{2(\text{hex.})}$ of 1127 K [18] are given for reference. The duration of each stage is explained in more detail below.

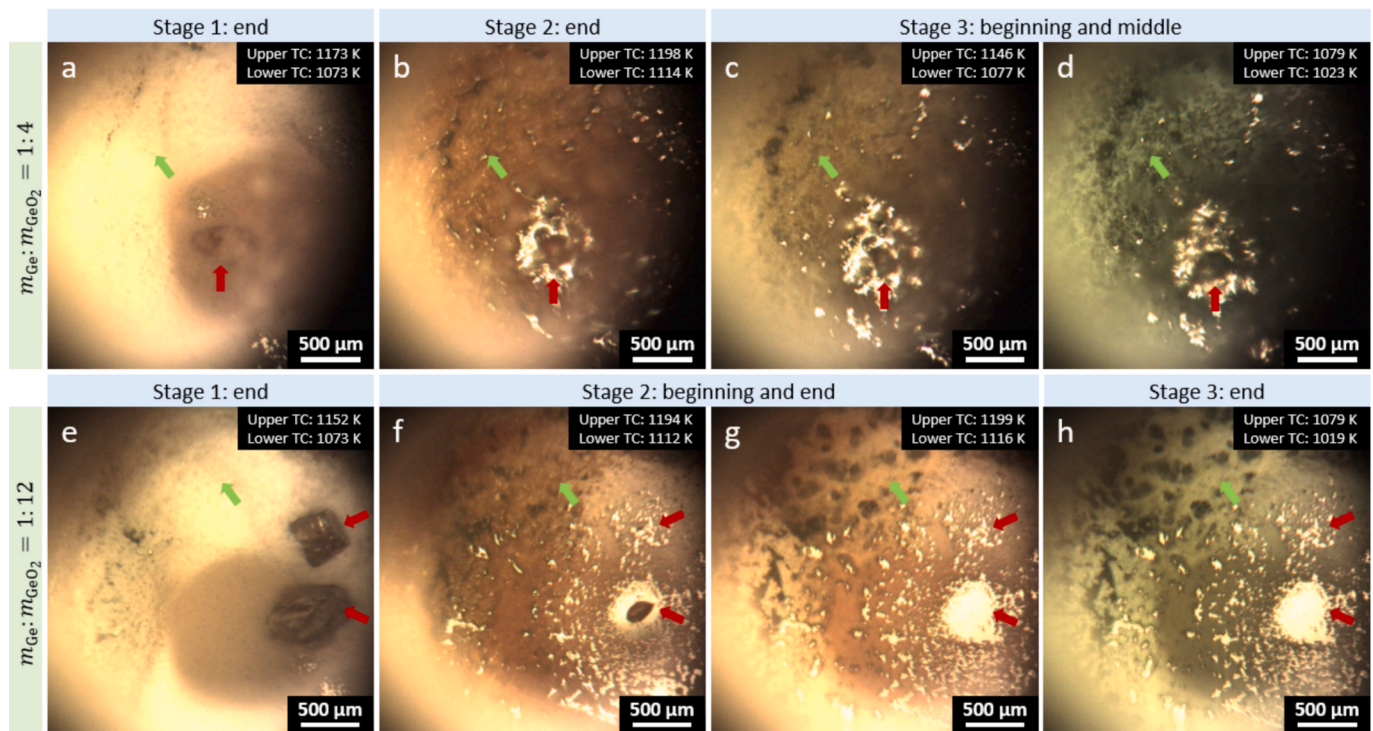


Fig. 3. In situ microscopy of the processes in the cell with reagents in the contact case during stages 1-3: **a-d** mass ratio of 1:4 and **e-h** 1:12. The time after the beginning of the first stage for images **a-d** is 739 s, 1831 s, 2042 s, 2879 s. For images **e-h**, this is 693 s, 2398 s, 4498 s, 4916 s respectively. The red arrow indicates the position of Ge splinters. The green arrow points to the same area on the GeO_2 powder so that one can trace the processes involved during all stages. The reddish colour at high temperatures is due to the thermal radiation of the BN TC plates.

During the first stage, the cell is heated to a temperature of 1073 K, at which it is completely sealed. It takes 10 to 15 min to reach this temperature. Fig. 3 a and e show cells at the end of the first stage.

Stage 2: formation of a liquid Ge- GeO_2 alloy

The second stage starts at 1073 K. The melting point of Ge equals 1211 K and of the hexagonal GeO_2 is at 1389 K [21]. The existence of the eutectic point in the Ge- $\text{GeO}_{2(\text{hex.})}$ system at 1127 K [18] explains melting point depression of GeO_2 powder and formation of a liquid alloy on its surface, which is clearly recognisable in the areas around the green arrows in Fig. 3 b and f. In most experiments, melting occurred at

Table 1

Parameters during the second stage. The first value in the range is the temperature at the beginning of the second stage, and the second one is at its end. During the second stage, there is a slightly increase in temperature, as can be seen in Fig. 2.

$m_{\text{Ge}} : m_{\text{GeO}_2}$	$T_{\text{upper TC}} [\text{K}]$	$T_{\text{lower TC}} [\text{K}]$	$\Delta t_{\text{stage 2}} [\text{min}]$
1:4	1190–1205 > $T_{\text{EP, hex.}}$	1105–1115 < $T_{\text{EP, hex.}}$	6–9
1:12	1195–1215 > $T_{\text{EP, hex.}}$	1095–1115 < $T_{\text{EP, hex.}}$	40–65
1:12	1225–1250 > $T_{\text{MP Ge}}$	1135–1175 > $T_{\text{EP, hex.}}$	< 5 / 5–9 / > 9

Notes: $m_{\text{Ge}} : m_{\text{GeO}_2}$ – the mass ratio of Ge to $\text{GeO}_{2(\text{hex.})}$; $T_{\text{upper TC}}$ and $T_{\text{lower TC}}$ – the average temperature of the upper and lower FS plates respectively; $T_{\text{EP, hex.}} = 1127 \text{ K}$ [18] – the eutectic point of the Ge- $\text{GeO}_{2(\text{hex.})}$ system; $T_{\text{MP Ge}} = 1211 \text{ K}$ [21] – the melting point of Ge; $\Delta t_{\text{stage 2}}$ – the duration of the second stage.

temperatures above 1070 K, but the lowest temperature at which melting of the alloy was observed is 1043 K. Considering the error, the temperature on the upper FS plate corresponding to this value equals $1131 \pm 8 \text{ K}$. The eutectic point is exactly within this range. As already discussed in our previous study [31], the alloy formation and its melting are related to the transfer of Ge to the GeO_2 powder, which appears to take place through the gas phase. Ge is transformed into a gaseous (g) state, either through sublimation of solid (s) Ge splinters, or when they are partly molten, also through vaporisation of liquid (l) Ge according to the following reaction:



Reaction (1) causes a reduction in the size of the Ge splinters. This is most noticeable in the contact case at the positions marked with red arrows in Fig. 3 b and f. Holes are formed in the powder around the splinters, through which light from the illumination below passes, as is noticeable when comparing, e.g., Fig. 3 e and f. The change in the shape of the powder may be attributed, firstly, to the formation of a liquid alloy and, secondly, to the following reaction, through which GeO(g) is produced, as already discussed in [31]:



At the beginning of the second stage, the alloy formation occurs. The further course of the reaction depends on the $m_{\text{Ge}} : m_{\text{GeO}_2}$ value. At a mass ratio of 1:4, the liquid alloy amount is a representative indicator of whether the nanoneedles, which are discussed in more detail in the next chapter, can be found in the cell after the third stage: if the liquid alloy amount is too much or too little, either no or very few needles are present in the cell. But this turns out to be known only during the ex situ analysis. However, the alloy formation can be observed via in situ microscopy, and its amount can be correlated with the results of ex situ analysis. It is necessary to end the second stage, when the whole powder is covered with a thin layer of the liquid alloy, as exemplarily depicted in Fig. 3 b, and proceed to the third stage. As can be seen from Table 1, the duration of the second stage, $\Delta t_{\text{stage 2}}$, for this mass ratio is between 6 and 9 min.

During the second stage, the size of Ge splinters decreases due to the formation of Ge(g) , according to reaction (1). In the case of $m_{\text{Ge}} : m_{\text{GeO}_2}$ equal to 1:4, it is not completely transformed into a gaseous state (Fig. 3 b). At a mass ratio of 1:12, 20–30 min into the second stage, almost nothing remains of the Ge splinter in the entire volume observable by in situ microscopy. Immediately or a few minutes after this, the liquid alloy starts to solidify and white regions are formed, as indicated, for example, by the green arrow in Fig. 3 f and g. After the completed formation of these regions, which can take up to 20–35 min, no further reactions are observed in the cell and the third stage can follow. The total duration, i. e., $\Delta t_{\text{stage 2}}$, at this mass ratio is between 40 and 65 min, as listed in Table 1. After interrupting the experiment just after the beginning of the formation of white regions and opening the cell, no Ge splinters can be found in its entire reaction chamber volume, even in the areas which in

situ microscopy cannot access. Therefore, it can be assumed that a necessary condition for the alloy to start solidifying and forming white regions is the complete sublimation or vaporisation of pure Ge splinters.

As discussed in the next chapter, during ex situ analysis of the cells, nanoneedles can only be detected in the white regions. In both methods described above, $T_{\text{lower TC}}$ is always below $T_{\text{EP, hex.}}$ and $T_{\text{upper TC}}$ is above $T_{\text{EP, hex.}}$. No noticeable difference in the cells with contact and non-contact reagent arrangements could be detected. The parameters such as $T_{\text{upper TC}}$, $T_{\text{lower TC}}$ and $\Delta t_{\text{stage 2}}$ for each $m_{\text{Ge}} : m_{\text{GeO}_2}$, summarised in the first two rows of Table 1 and exemplarily represented in Fig. 2, can be considered as optimal conditions for growing a large number of nanoneedles. The latter, shown for example in Fig. 4 b and c, is the main objective of this study.

At $T_{\text{lower TC}}$ above $T_{\text{EP, hex.}}$, less nanoneedles and more crystals with other morphologies can be found in the cell, as shown in Fig. 4 d-i. In this case, $T_{\text{upper TC}}$ is higher than not just $T_{\text{EP, hex.}}$, but also $T_{\text{MP Ge}}$. However, additional consideration of the experimental results beyond optimal conditions contributes to a more detailed understanding of how nanoneedles grow. The third row of Table 1 presents the parameters in these experiments. For $\Delta t_{\text{stage 2}}$ between 5 and 9 min, the amount of alloy is about the same as that depicted in Fig. 3 b. If the duration is shorter or longer, a deficiency or excess of alloy is formed respectively. The results of all these three cases at $T_{\text{lower TC}}$ above $T_{\text{EP, hex.}}$ are discussed in the next chapter.

Stage 3: cooling of the cell

In the third stage, the cell is cooled down. The cooling rate of both FS plates is 0.1–0.5 K/s in all experiments. In the case of $m_{\text{Ge}} : m_{\text{GeO}_2}$ of 1:4 and $T_{\text{lower TC}}$ below $T_{\text{EP, hex.}}$, at temperatures below approx. 1085 K, white regions begin to become visible (Fig. 3 c) and take their final appearance (Fig. 3 d) in some cases down to 850 K, at which the temperature difference between the upper and lower FS plates is only up to 40 K. In parallel with the formation of white regions, dark grey regions are also formed directly next to them. At a mass ratio equal to 1:12, the formation of these regions during the third stage is not observable. No noticeable differences between the situation in the cell at the end of the second phase and at the end of the third phase can be detected (Fig. 3 g and h). The only marked contrast is the reddish colour, which, however, is due to the thermal radiation of the BN TC plates at high temperatures.

When cooling down from temperatures above $T_{\text{EP, hex.}}$, the same processes occur as in the first case described above. The separation into white and dark grey regions is also observed here.

3. Results of ex situ analysis

After cooling down, when 573 K is reached, the vacuum system is turned off. Once ambient pressure and temperature inside the vacuum chamber is established, the cell can be removed from it. The circular notch in one of the FS plates (Fig. 1) serves as a crack line when mechanical stress is applied. The cell can be opened easily by slightly breaking a small hole in the notch with the tip of a tweezers, then picking up and taking out a small platform-like area with the powder residues and reaction products. Subsequently, it is placed on an SEM holder, coated with amorphous carbon, and analysed with FE-SEM (JEOL JSM-6700F) as well as EDX spectroscopy (Oxford Instruments INCA 300). The results are presented in Figs. 4 to 7. For Raman spectroscopy (Olympus BX microscope with Raman spectrometer Bruker Senterra), the samples remained uncoated, as shown in Fig. 5 a. The needles from samples, coated and uncoated with carbon, are placed on a TEM grid and analysed with for FE-TEM (JEOL JEM-2100F-UHR). The results are shown in Fig. 8. Bright-field and dark-field TEM, HRTEM images and SAED patterns are taken, and EDX spectroscopy (Oxford Instruments INCA 200) is performed.

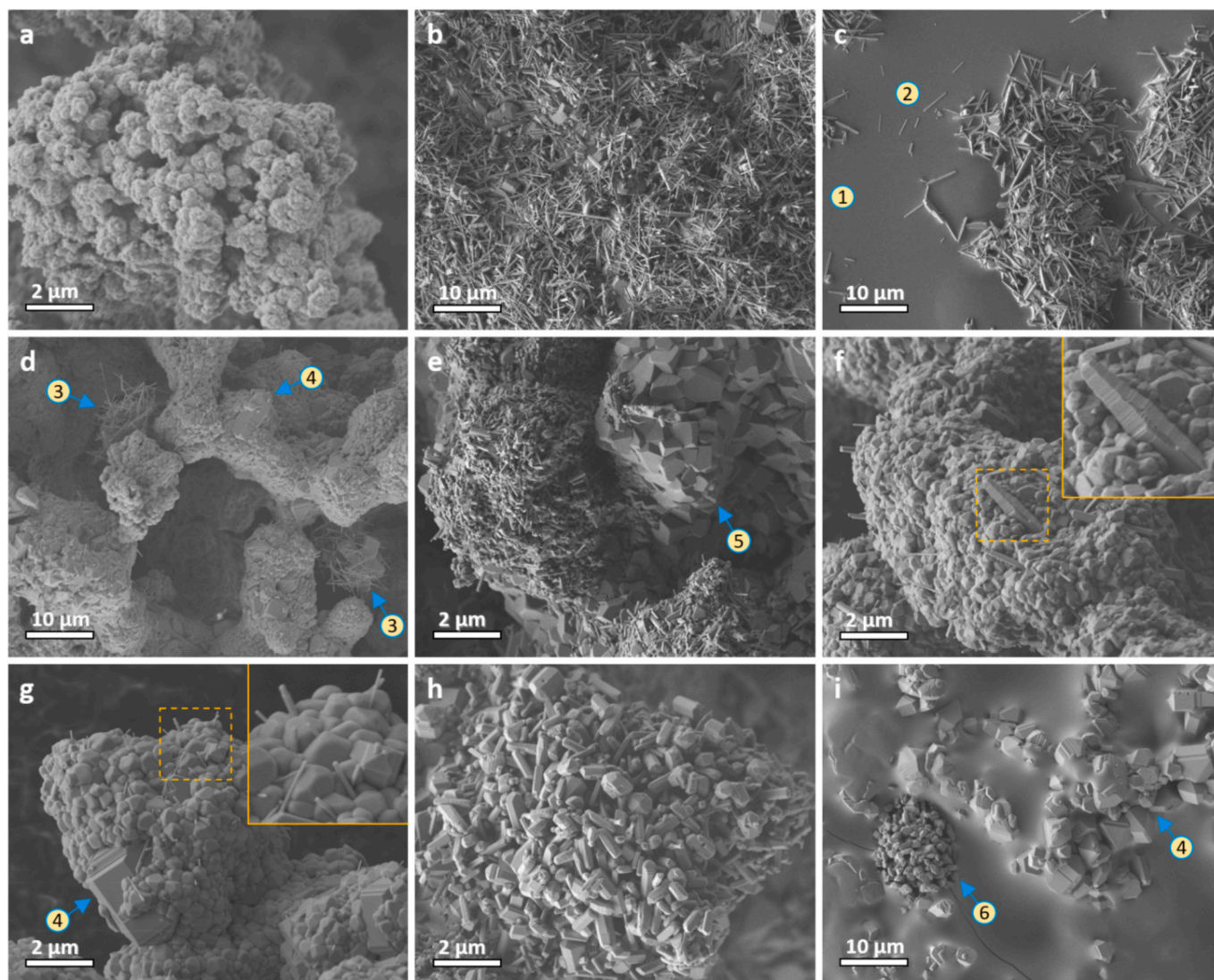


Fig. 4. SEM images of powder **a** before the heating and cooling procedures and after at $T_{\text{lower TC}}$ during the second phase **b**, **c** below and **d**–**i** above $T_{\text{EP, hex.}}$. Numerical designations (1)–(6) are explained in the text.

3.1. Analysis of reaction products in the chemical transport cell

For comparison, Fig. 4 **a** shows the GeO_2 powder as it is before the heating and cooling procedures, and Fig. 4 **b**–**i** depict the structures in the cell after the procedures. Both methods for growing nanoneedles at $T_{\text{lower TC}}$ below $T_{\text{EP, hex.}}$ during the whole second stage (first two rows in Table 1) show almost identical results: a large number of nanoneedles with thicknesses of 50–500 nm and lengths of 5–7 μm , as shown in Fig. 4 **b** and **c**. Almost all needles are located on the powder. On the FS plate, they are either absent (1) or there are only few separately lying needles (2).

At $T_{\text{lower TC}}$ above $T_{\text{EP, hex.}}$, less nanoneedles and more larger crystals with other morphologies are present, as depicted in Fig. 4 **d**–**i**. When the duration of the second stage is between 5 and 9 min (last row of Table 1), the powder is covered with a rather homogeneous layer of crystals, an example of which is shown in Fig. 4 **h**.

At durations of less than 5 min, many bare regions are left on the powder, where only crystals that have been fully fused together can be found. An example of such a region is shown in Fig. 4 **e** and labelled with (5). As can be seen in the same Fig. 4 **e**, next to this region, there may be other regions in which the density of the nanoneedles and crystals with other morphologies is quite high. From the analysis of such cells, the

following summary can be made: single needles can either be assembled into aggregates with a tumbleweed-like form (3), which are also shown on a larger scale in Fig. 5 **c** and Fig. 6 **g**, or they can lie on the powder (Fig. 4 **e**) or grow from it outwards (Fig. 4 **f** and **g**). The polyhedral crystals can be submicrometric (6), to almost 10 μm (4) in size.

When the cell is heated at $T_{\text{lower TC}}$ above $T_{\text{EP, hex.}}$ for more than 9 min, a large amount of the liquid alloy is formed. The solidified alloy is shown in Fig. 4 **i**.

Fig. 5 **a** shows an ex situ optical microscope image of the powder surface after the experiment. White (position 1) and dark grey regions (position 2) are clearly distinguishable in the image. Their formation can be observed via in situ microscopy, as shown in Fig. 3 **d**. Both regions are detailed in Fig. 5 **b**. Small crystals (positions 3 and 4), including the nanoneedles, can only be found in white regions. The dark grey regions are composed of large crystals (positions 6 and 7), including those that are fused together, which are labelled in Fig. 4 **e** as (5). These are the only regions that have the solidified alloy, as shown in Fig. 4 **i**.

Dark grey regions are formed mainly at the parameters during the second stage stated in the first and third rows of Table 1. Under certain conditions during the second stage, e.g., $T_{\text{lower TC}}$ above $T_{\text{EP, hex.}}$ and $\Delta t_{\text{stage 2}}$ more than 9 min, their number can be quite large, as shown in Fig. 5. When growing nanoneedles with parameters from the second row

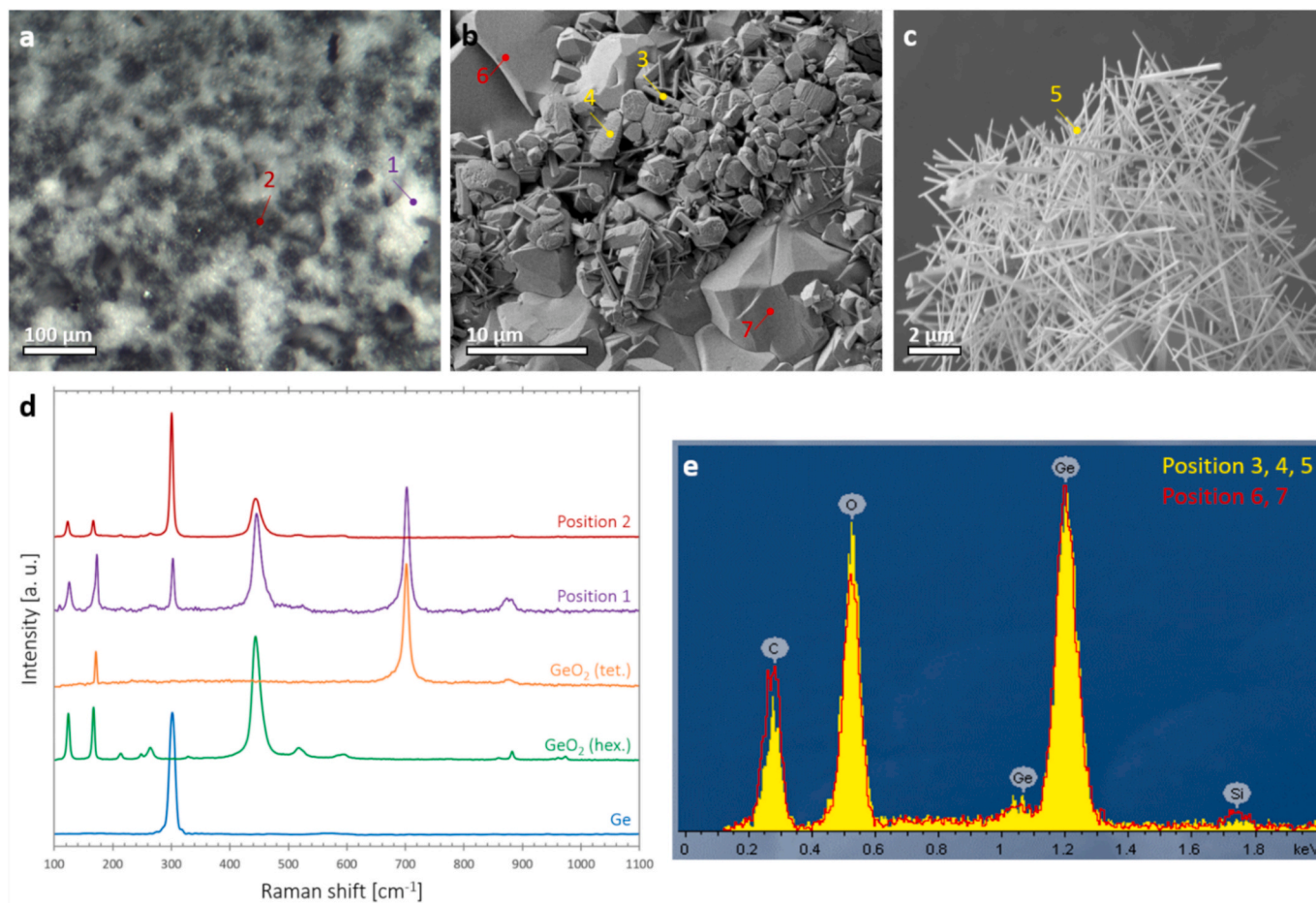


Fig. 5. **a** Ex situ optical microscope image of the powder (position marked with green arrow in Fig. 3 d) and **b, c** SEM images of some positions on the powder after the whole heating and cooling procedure. **d** Raman spectra at positions 1 and 2, and of the reference spectra of pure substances – crystalline Ge, hexagonal GeO₂ and tetragonal GeO₂. **e** EDX spectra at positions 3–7.

of Table 1, grey regions are almost completely absent. In this case, the powder appears white, as shown in Fig. 3 h, and is almost completely covered with nanoneedles.

The Raman spectroscopy (Fig. 5 d) reveals differences in composition in the white and dark grey regions: while in position 2, there are peaks of crystalline Ge and hexagonal GeO₂ only, in position 1 there are also additional peaks of tetragonal GeO₂. It can be assumed that the cause of the latter peak is the small crystals, including the nanoneedles, which can only be found in the white regions. The reference spectra from our measurements of the pure substances, crystalline Ge, hexagonal GeO₂ and tetragonal GeO₂, are also plotted in Fig. 5 d and match the spectra from other sources, e.g., [15,34] and [35].

The results of the SEM EDX spectroscopy with acceleration voltage of 5 kV at positions 3–7 are shown in Fig. 5 e. The most significant difference between fine (positions 3–5) and large crystals (positions 6 and 7) is that the latter contain more Si. To quantify the ratio of atomic percentages of O to Ge, at%[O]:at%[Ge], at these positions, the EDX measurements were calibrated with a pure hexagonal GeO₂ powder. It remains unclear whether Si is incorporated either as elemental Si or as a chemical compound, e.g., SiO₂. Two estimates can therefore be made: with Si and with SiO₂ as impurity. For positions 3–5, at%[O]:at%[Ge] is 1.75 (Si as impurity) and 1.73 (SiO₂ as impurity). The atomic percentage of Si, at%[Si], itself is less than 1 %. For positions 6 and 7 with Si and with SiO₂ as impurity, at%[O]:at%[Ge] equals 2.03 and 1.70 respectively. Here, at%[Si] is up to 4 %.

No Si peak is presumably present in the Raman spectrum of position 2 (Fig. 5 d), in which, according to the EDX spectroscopy, the Si content

of a few per cent is expected. There is a weakly visible peak on this spectrum between approx. 515 cm⁻¹ and 525 cm⁻¹. It is more likely to be related to hexagonal GeO₂ at 516 cm⁻¹ [15] than to crystalline Si at 520 cm⁻¹ [36]. This assumption can be supported by the fact that the position of this peak coincides with that of the peak on the spectrum of hexagonal GeO₂.

In order to study the composition of the alloy shown in Fig. 4 i separately from the powder lying underneath it, at $m_{\text{Ge}} : m_{\text{GeO}_2}$ of 1:4 and $T_{\text{lower TC}}$ above $T_{\text{EP, hex.}}$ the second phase can be prolonged until all the powder in the cell is fully utilised to form a liquid alloy. Such an alloy is transparent and yellowish, as demonstrated in our previous study [31], but turns dark brown after solidification. To minimise the influence of the FS plate underneath the alloy on the results, a section of solidified alloy with a thickness of at least 20 μm is selected for the analysis. The Raman spectroscopy reveals only one peak in the solidified alloy, which corresponds to a single peak of crystalline Ge at approx. 300 cm⁻¹, as plotted in Fig. 5 d. The SEM EDX spectroscopy shows that in this alloy, the ratio at%[O]:at%[Ge] is equal approx. to 1.40 and 1.14, with Si and with SiO₂ as impurity, respectively. at%[Si] is between 3.5 % and 7 %.

Comparing the alloy layer on the powder (dark grey regions – position 2) and on the FS plate, the following conclusions can be made. In both cases, the qualitative composition is identical and includes elements O, Ge and Si, in addition to crystalline Ge being detected in both. It can be hypothesised that the alloy layer on the powder is thinner because, firstly, it has a higher at%[O] and, secondly, the Raman spectroscopy also detects the peaks of hexagonal GeO₂, which are absent in

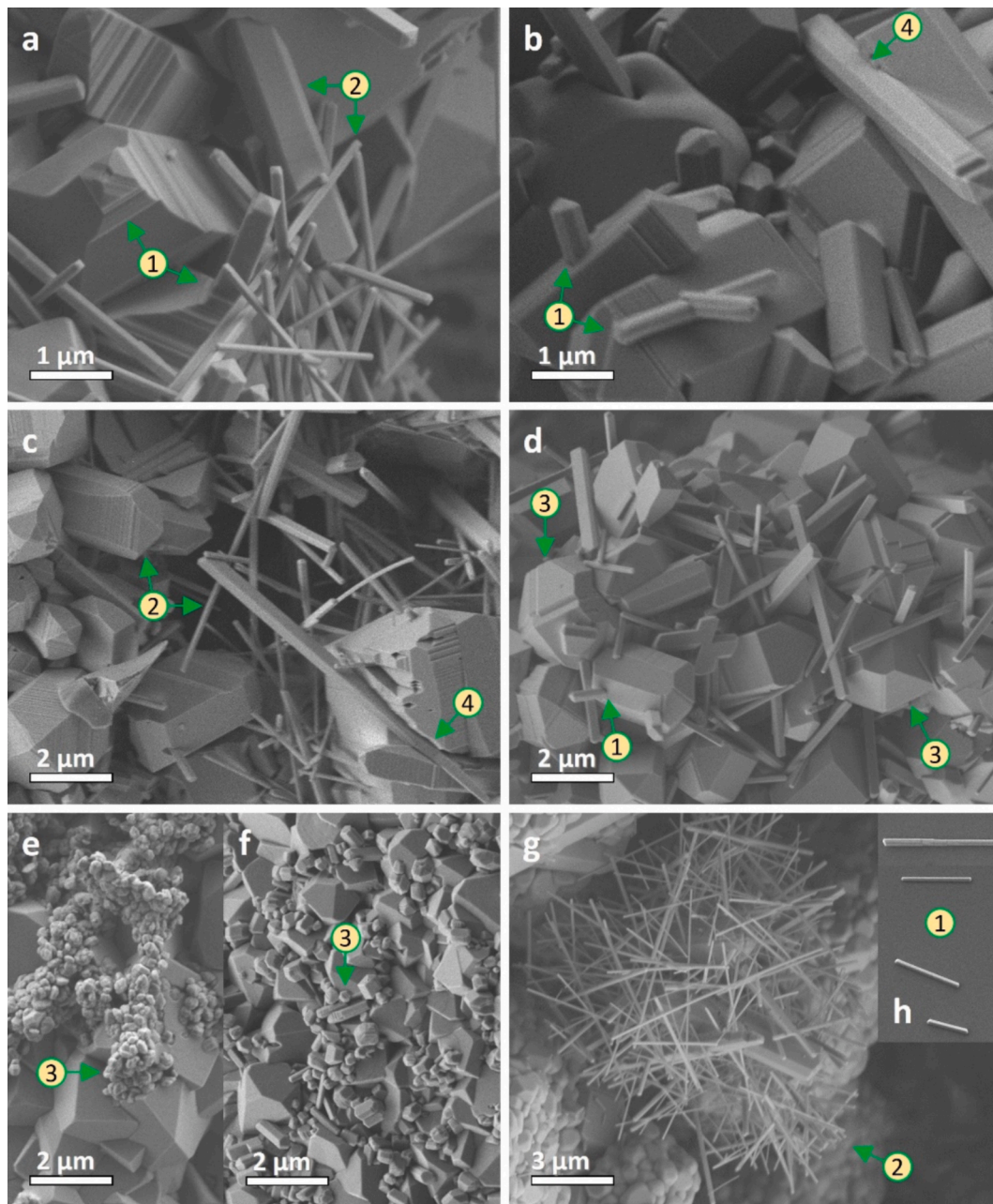


Fig. 6. Morphology of crystals formed on the powder after the heating and cooling procedure. Details about **a–h** as well as an explanation of numerical designations (1)–(4) can be found in the text. Scale in **g** and **h** is the same.

the alloy on the FS plate. Si in both samples probably comes from the Corning® EAGLE XG® glass spacer, which turns soft at high temperatures, or can be understood in terms of weak reactions of Ge and GeO₂ with the FS plates.

3.2. Analysis of crystal morphology

Fig. 6 shows the morphology of crystals formed on the powder surface. The crystals with acicular habit (needles) and prismatic habit, as indicated by (1), can grow out of crystal faces, i.e., have only one growing end (Fig. 6 **a** and **b**), or grow in both directions in the longitudinal axis (Fig. 6 **h**). In the latter case, they can either just lie on the powder surface (Fig. 4 **b** and **c**) or be assembled in aggregates with a tumbleweed-like form (Fig. 5 **c** and Fig. 6 **g**). In both cases, however, they can be separated from the powder underneath quite easily. The needles and prismatic crystals can be of different thicknesses (2):

approx. 50–100 nm, which are suitable for the TEM analysis, up to approx. 5 μm (Fig. 7 **d**), and with thicknesses with values between these two limits. At $T_{\text{lower TC}}$ above $T_{\text{EP, hex}}$, polyhedral crystals (3) predominate. Their sizes can also vary from several hundred nanometres (Fig. 6 **e** and **f**) and a couple of micrometres (Fig. 6 **d**) to almost 10 μm (Fig. 4 **i**). The crystals can lie heavily clumped together or be quite uniformly distributed over the surface, as shown in Fig. 6 **e** and **f** respectively.

After opening the cell, many needles can be found lying separately, as shown in Fig. 4 **c**, from which it can be assumed that they are also easily separable from each other. Fig. 6 **g** shows that in the aggregates with a tumbleweed-like form, the nanoneedles are bonded together only in a small contact area. This is shown in more detail in Fig. 8 **a**. The contact areas (4) are clearly visible on a micrometre scale in Fig. 6 **b** and **c**.

Further discussion is focused on the needles (Fig. 7 **b**) and prismatic crystals (Fig. 7 **c** and **d**). Fig. 7 **a** presents three crystal models based on

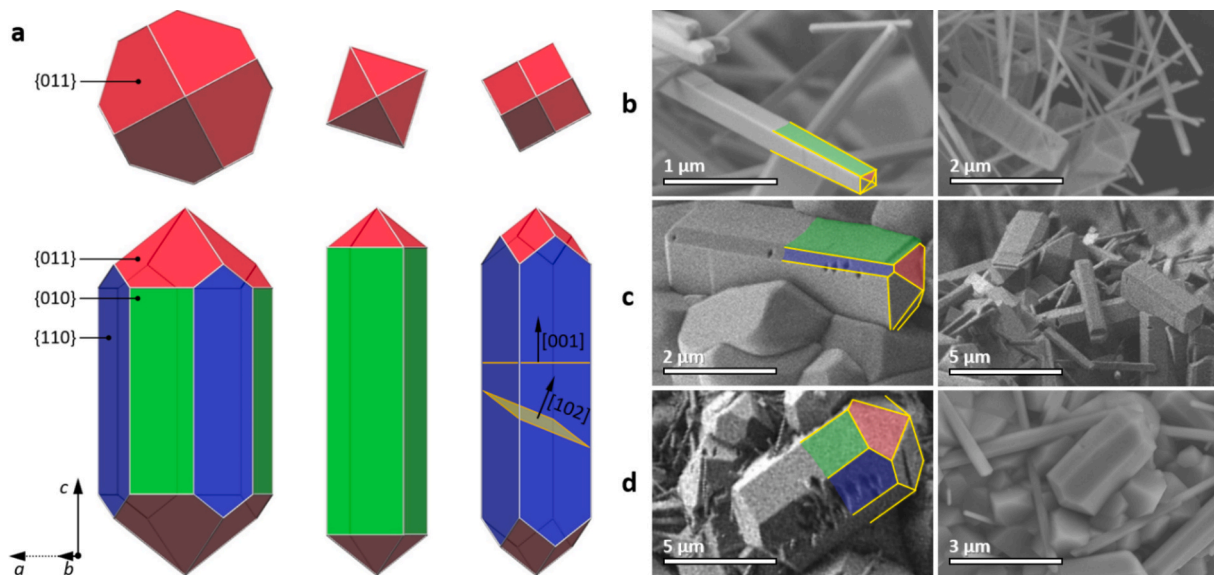


Fig. 7. a Model of tetragonal crystals with a different tract: ditetragonal prismatic (left) and tetragonal prismatic (centre, right) body with tetragonal pyramidal (centre) or trapezohedral (right) tip, or a combination of these two (left). The top row is a top view of the crystals; the bottom row is a side view. The red, green and blue colours indicate the crystal faces corresponding to planes $\{011\}$, $\{010\}$ and $\{110\}$, respectively. $[001]$ and $[102]$ indicate directions from Fig. 8. b-d Examples of real crystals with different tract.

the morphology of real crystals found on SEM images, and the Raman spectroscopy results and made with the software KrystalShaper [37]. They have a rutile-like structure and correspond to a tetragonal crystal system. As can be seen, the body of these crystals is prismatic and their tips can be pyramidal, trapezohedral, or a combination of the two. Both on the models itself and on the real crystals, the red, green and blue colours indicate crystal faces corresponding to planes $\{011\}$, $\{010\}$ and $\{110\}$, respectively. The crystal tract is primarily determined by the degree of development of the latter two prism faces. Real crystals corresponding to the first two models are shown in Fig. 7 b-d. The third model describes the morphology of the nanoneedle depicted in Fig. 8 b-f.

3.3. Analysis of nanoneedles

Fig. 8 a shows nanoneedles on a TEM grid. Dark-field TEM images (Fig. 8 b and c) confirm that they are single crystals. Both dark-field and bright-field (Fig. 8 d) TEM images show a needle fragment, its SAED pattern obtained through experiments is shown in Fig. 8 f. The CarIne Crystallography software [38] and CIF (Crystallographic Information File) with a model of tetragonal GeO_2 (ICSD 92551), based on [39], are used to simulate the SAED patterns. Comparison of the experimental pattern (Fig. 8 f) and the simulated pattern reveals that the former corresponds to the zone axis $[110]$. From the spot indexing of this zone axis, it can be concluded that the longitudinal growth of nanoneedles occurs in the $[001]$ direction, i.e., coincides with the c -axis, as is depicted in Fig. 7 a. In Fig. 8 d, the orange arrow indicates the location at which the HRTEM image represented in Fig. 8 e is made. In the latter image, the values of the spacing between lattice planes in the longitudinal and lateral directions are shown, as well as the direction of the needle growth discussed above. According to the HRTEM analysis, it can be hypothesized that the surface of the needles is not covered with an amorphous multilayer, since families of lattice planes extend to the periphery, as can be seen, e.g., at the very tip of the needle in Fig. 8 e.

The same procedure is carried out with another nanoneedle coated with amorphous carbon (Fig. 8 g-i), which is tilted at a different angle than the first nanoneedle. Fig. 8 i shows the SAED pattern corresponding to the zone axis $[21\bar{1}]$, from which the longitudinal direction of needle is $[102]$. The relative orientation of $[102]$ and the longitudinal needle

growth direction $[001]$ is represented in Fig. 7 a.

The TEM EDX spectroscopy with acceleration voltage of 40 kV reveals that at%[Si] in both nanoneedles in Fig. 8 is less than 0.4 %.

4. Discussion

From the results provided above, it can be concluded that the nanoneedles are composed of GeO_2 and have a tetragonal crystal structure. This follows independently from the comparison of experimentally obtained SAED patterns (Fig. 8 f and i) with simulated SAED patterns, and from the combination of the Raman spectroscopy, optical microscopy and SEM results (Figs. 4 and 5). The conclusion regarding the crystal structure is also confirmed by comparing the morphology of nanoneedles (Fig. 8 b-d) with the constructed model of a tetragonal crystal (Fig. 7 a). When the parameters of the model are varied, its tract changes, corresponding to the real crystals that can be found by the SEM analysis of the samples (Fig. 7).

It can be concluded from the TEM analysis that the nanoneedles grow in the $[001]$ direction (Fig. 8 e and f) and they are single crystals without an amorphous multilayer on their surface (Fig. 8 b, c and e). This direction of growth agrees with the direction proposed in [11] and [28] for rutile GeO_2 rods, based on experimental data and calculations. Planes $\{011\}$, $\{010\}$ and $\{110\}$ belong to the set of most stable surfaces in tetragonal GeO_2 , with the latter two having the lowest surface energy among all others [28]. This agrees with the morphology of crystals from our experiments presented in Figs. 7 and 8. The prismatic and acicular growth habits of $\text{GeO}_{2(\text{tet.})}$ also match the frequently formed habits of rutile [40,41]. The values of spacings between lattice planes (Fig. 8 e) coincide well with the values from [22] for $\text{GeO}_{2(\text{tet.})}$ substrate oriented along the zone axis $[110]$.

The following model can be hypothesised to explain the formation and growth of the tetragonal GeO_2 nanoneedles. At high temperatures, starting from the end of the first stage, Ge(g) is produced according to reaction (1). In the temperature range from 1043 K to 1215 K, the latter being the maximum temperature in the cell during the second stage (Table 1), the partial pressure of Ge(g) is equal to $(0.01\text{--}1.54) \times 10^{-6}$ mbar, according to our thermodynamic calculations using the software Thermodynamik [42], as well as data from [43]. Ge(g) then reacts with the hexagonal GeO_2 powder, with which it forms a liquid eutectic alloy. The metastable eutectic is characterised by $T_{\text{EP, hex.}} = 1127$ K and a GeO_2

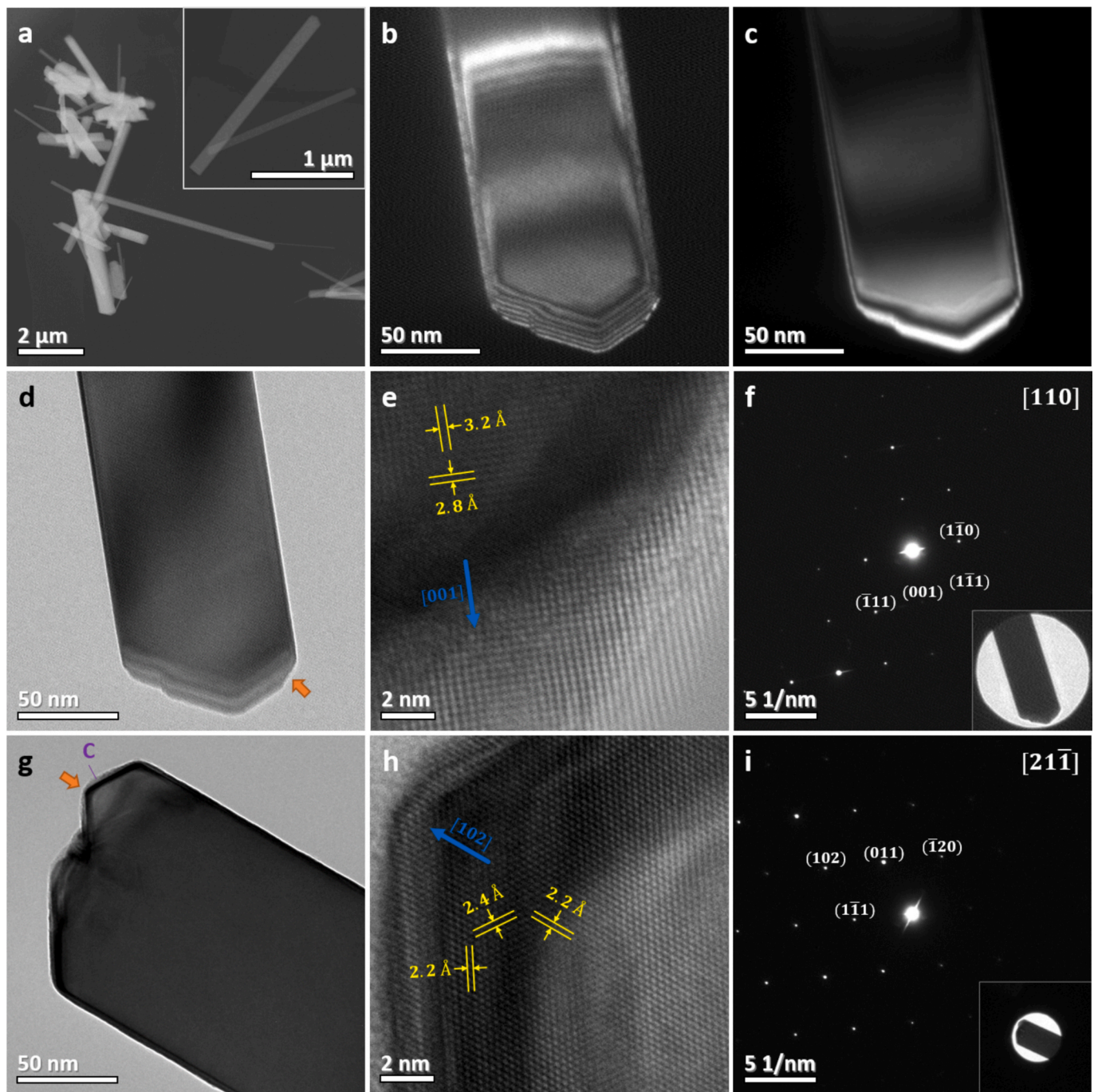


Fig. 8. TEM analysis of the nanoneedles. **a** Needles on a TEM grid. The inset shows separately arranged needles in a magnified scale. Two nanoneedles, designated in the text as needle 1 (uncoated) and needle 2 (coated with amorphous carbon – labelled as C in **g**), which are shown in **b–f** and **g–i** respectively, are selected for a more detailed study. **b, c** Dark-field and **d, g** bright-field TEM images recorded from these needles. **e, h** show HRTEM images taken for two different tilt angles at the tips of needles 1 and 2 (locations at which these high-resolution images were made are indicated by orange arrows in **d** and **g**). **[001]** and **[102]** are directions of corresponding crystal planes of needle 1 and 2 respectively (their relative direction to each other is shown in Fig. 7). **f, i** SAED patterns in **[110]** and **[211]** zone axes.

mole fraction of $x_{\text{GeO}_2}^{\text{EP, hex.}} = 0.765$ [18]. According to our observations, the alloy formation occurs shortly before the beginning of the second stage at $T_{\text{lower TC}}$ of at least 1043 K, which corresponds to $T_{\text{upper TC}}$ of 1131 K. Considering the temperature error, $T_{\text{lower TC}}$ remains at least 5 K below $T_{\text{EP, hex.}}$ throughout the second stage. Under such conditions, presumably mainly at the top of the powder, are the conditions under which the liquid alloy can be formed. This process occurs slowly, allowing the amount of the alloy to be controlled by in situ microscopy quite well (Fig. 3).

Further course of reactions in the cell depends on the mass ratio of Ge

to $\text{GeO}_2(\text{hex.})$. At $m_{\text{Ge}} : m_{\text{GeO}_2}$ of 1:12, which corresponds to $x_{\text{GeO}_2} = 0.893$, after a few tens of minutes, it comes to a point when there are no more Ge splinters left in the cell. Even assuming that all of the Ge reacts with the powder through the gas phase, in this case, x_{GeO_2} is still greater than $x_{\text{GeO}_2}^{\text{EP, hex.}}$, which means that a part of the powder remains solid, but covered by the liquid alloy. Besides the metastable eutectic, there is also a stable eutectic in the Ge-O phase diagram, which is at $T_{\text{EP, tet.}} = 1189.4$ K and $x_{\text{GeO}_2}^{\text{EP, tet.}} = 0.71$ [18]. After the Ge(g) supply is stopped, the stable tetragonal phase starts to form. At temperatures below $T_{\text{EP, tet.}}$,

solidification and decomposition of the alloy into Ge(s) and GeO_{2(tet.)} occurs according to the phase diagram in [18]. During this process, nuclei of the nanoneedles are presumably formed, and Ge(s) is distilled out and again transferred to the gas phase, as the temperature remains high enough during the whole second stage. This also takes several tens of minutes. These processes are not directly observable, but the gradual disappearance of the liquid alloy and the emergence of white regions, which consist of the nanoneedles, can well be followed by in situ microscopy, as shown in Fig. 3 f and g. During the third stage, when the cell cools down, the formation of the dark grey regions is not observable, which may indicate that all the Ge has already sublimated during the decomposition of the alloy in the second stage, as discussed above.

In the case $m_{\text{Ge}} : m_{\text{GeO}_2}$ of 1:4, i.e., $x_{\text{GeO}_2} = 0.735$, which is close to $x_{\text{GeO}_2}^{\text{EP, hex.}}$, the second stage is to be stopped before all the powder completely reacts with Ge(g) to form a liquid alloy. The decomposition of the alloy during cooling is observable via in situ microscopy (Fig. 3 c), starting from $T_{\text{lower TC}}$ of about 1085 K, with $T_{\text{upper TC}}$ being approx. 1155 K, which is lower than $T_{\text{EP, tet.}}$. At these temperatures in the cell, the partial pressure of Ge(g) reaches 0.22×10^{-6} mbar at most [42,43], and continues to drop with decreasing temperature. Therefore, Ge(s) that is formed during decomposition is not sublimated as it was at higher temperatures using the other method of growing nanoneedles described above. This could probably explain the formation of a large number of dark grey regions, in which Ge content is quite high, along with white regions with nanoneedles. Based on the results of the EDX and Raman spectroscopy, the dark grey regions may be made of a solidified alloy that is composed of amorphous germanium oxide and crystalline Ge. This alloy covers the hexagonal GeO₂ powder.

In summary, it can be speculated that the formation of needles is related to the separation of the stable tetragonal phase from the liquid alloy during its decomposition. The transport of Ge(g) to the powder surface suppresses decomposition, and therefore the latter is possible either in the case of complete depletion of Ge source, i.e., splinters, with continued heating at $m_{\text{Ge}} : m_{\text{GeO}_2}$ of 1:12, or during cooling of the cell, when the partial pressure of Ge(g) decreases rapidly, as in the case of $m_{\text{Ge}} : m_{\text{GeO}_2}$ of 1:4.

During the second stage, significant quantities of gaseous GeO are produced through reaction (2). According to our thermodynamic calculations using [42] and data from [43] and [44], the partial pressure of GeO(g) is the highest among those of other gases in the cell. It is equal to approx. 120 mbar at 1215 K. During the cooling of the cell in the third stage at $T_{\text{lower TC}}$ below 990 K, with $T_{\text{upper TC}}$ being below 1050 K, formation and growth of composite germanium/germanium oxide micro-needles occur, which could be related to the strong deposition of GeO and its complete disproportionation into Ge and germanium oxide, as discussed in more detail in our previous study on microneedles [31]. Therefore, it may be assumed that in both nanoneedle growth methods, GeO(g) is present in the cell. However, it remains unclear whether and how GeO is involved in processes of nuclei formation and growth of the tetragonal GeO₂ nanoneedles. However, one might speculate that GeO(g) could be a source of GeO₂ for the needles according to the following disproportionation:



The formation of tumbleweed-like aggregates that rise above the powder surface by more than 10 μm (Fig. 5 c) might be explained by the involvement of the reaction (3) in the nanoneedle growth process.

5. Conclusions

Two high-temperature methods for growing tetragonal GeO₂ nanoneedles were presented. Processes throughout all stages of the methods were observed in situ by means of a video microscopy. Reaction products were analysed ex situ with SEM, EDX and Raman spectroscopy. The crystal structure of the nanoneedles was studied with TEM, and their

longitudinal growth direction was also determined. The morphology of crystals, including nanoneedles, was discussed and their temperature dependence was shown. A model for the formation of nanoneedles was also proposed.

CRedit authorship contribution statement

Alexey S.T. Rybakov: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Vu Anton Lie:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Louis Mirecki:** Investigation. **Jörg August Becker:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Prof. Dr. Armin Feldhoff from the Institute of Physical Chemistry and Electrochemistry (PCI), Leibniz University Hannover (LUH), for providing us access to the FE-SEM (JEOL JSM-6700F) with the EDX spectrometer (Oxford Instruments INCA 300), the FE-TEM (JEOL JEM-2100F-UHR) with the EDX spectrometer (Oxford Instruments INCA 200) and the preparation laboratories, as well as for the support of our electron microscopy research. We are very grateful to Frank Steinbach from the PCI for carrying out all TEM measurements, for his strong involvement in analysing the samples as well as for his generous technical advice. We thank M.Sc. Merle Wellmann from the PCI for giving us an excellent introduction to the CaRIne Crystallography software and for her advice on analysing our samples. We also extend our thanks to Prof. Dr. Claus Rüschler from the Institute of Mineralogy, LUH for allowing us access to the microscope (Olympus BX) with the Raman spectrometer (Bruker Senterra) and for supporting our research.

Data availability

Data will be made available on request.

References

- [1] E. Roduner, *Nanoscale Materials: Size-Dependent Phenomena*, Royal Society of Chemistry, Cambridge, 2006.
- [2] X.C. Wu, W.H. Song, B. Zhao, Y.P. Sun, J.J. Du, Preparation and photoluminescence properties of crystalline GeO₂ nanowires, *Chem. Phys. Lett.* 349 (3–4) (2001) 210–214, [https://doi.org/10.1016/S0009-2614\(01\)01213-1](https://doi.org/10.1016/S0009-2614(01)01213-1).
- [3] P. Hidalgo, B. Méndez, J. Piqueras, GeO₂ nanowires and nanoneedles grown by thermal deposition without a catalyst, *Nanotechnology* 16 (11) (2005) 2521–2524, <https://doi.org/10.1088/0957-4484/16/11/010>.
- [4] Z. Gu, F. Liu, J.Y. Howe, M.P. Paranthaman, Z. Pan, Three-dimensional germanium oxide nanowire networks, *Cryst. Growth Des.* 9 (1) (2009) 35–39, <https://doi.org/10.1021/cg8010986>.
- [5] M. Peng, Y. Li, J. Gao, D. Zhang, Z. Jiang, X. Sun, Electronic structure and photoluminescence origin of single-crystalline germanium oxide nanowires with green light emission, *J. Phys. Chem. C* 115 (23) (2011) 11420–11426, <https://doi.org/10.1021/jp201884y>.
- [6] Y.H. Tang, Y.F. Zhang, N. Wang, I. Bello, C.S. Lee, S.T. Lee, Germanium dioxide whiskers synthesized by laser ablation, *Appl. Phys. Lett.* 74 (25) (1999) 3824–3826, <https://doi.org/10.1063/1.124192>.

- [7] Z. Jiang, T. Xie, G.Z. Wang, X.Y. Yuan, C.H. Ye, W.P. Cai, G.W. Meng, G.H. Li, L. D. Zhang, GeO₂ nanotubes and nanorods synthesized by vapor phase reactions, *Mater. Lett.* 59 (4) (2005) 416–419, <https://doi.org/10.1016/j.matlet.2004.09.036>.
- [8] S.L. Shinde, K.K. Nanda, Thermal oxidation strategy for the synthesis of phase-controlled GeO₂ and photoluminescence characterization, *CrstEngComm* 15 (6) (2013) 1043–1046, <https://doi.org/10.1039/C2CE26721G>.
- [9] K.N. Astankova, E.B. Gorokhov, I.A. Azarov, V.A. Volodin, A.V. Latyshev, Local anodic oxidation of solid GeO films: The nanopatterning possibilities, *Surf. Interfaces* 6 (2017) 56–59, <https://doi.org/10.1016/j.surfin.2016.11.010>.
- [10] Y.A. Bioud, E. Paradis, A. Boucherif, D. Drouin, R. Arès, Shape control of cathodized germanium oxide nanoparticles, *Electrochem. Commun.* 122 (2021) 106906, <https://doi.org/10.1016/j.elecom.2020.106906>.
- [11] C. Jing, W. Sun, W. Wang, Y. Li, J. Chu, Morphology and crystal phase evolution of GeO₂ in liquid phase deposition process, *J. Cryst. Growth* 338 (1) (2012) 195–200, <https://doi.org/10.1016/j.jcrysgro.2011.11.024>.
- [12] V.L. Deringer, M. Lumeij, R.P. Stoffel, R. Dronskowski, Ab initio study of the high-temperature phase transition in crystalline GeO₂, *J. Comput. Chem.* 34 (27) (2013) 2320–2326, <https://doi.org/10.1002/jcc.23387>.
- [13] Y. Tanaka, T. Takeuchi, S.W. Lovesey, K.S. Knight, A. Chainani, Y. Takata, M. Oura, Y. Senba, H. Ohashi, S. Shin, Right handed or left handed? Forbidden x-ray diffraction reveals chirality, *Phys. Rev. Lett.* 100 (2008) 145502, <https://doi.org/10.1103/PhysRevLett.100.145502>.
- [14] The AFLOW standard encyclopedia of crystallographic prototypes, Enantiomorphic Space Groups. <https://aflow.org/p/enantiomorph.info.html>, 2024 (accessed 13 June 2025).
- [15] M. Micoulaut, L. Cormier, G.S. Henderson, The structure of amorphous, crystalline and liquid GeO₂, *J. Phys.: Condens. Matter* 18(45) (2006) R753–R784, <https://doi.org/10.1088/0953-8984/18/45/R01>.
- [16] A.W. Laubengayer, D.S. Morton, Germanium. XXXIX. the polymorphism of germanium dioxide, *J. Amer. Chem. Soc.* 54 (6) (1932) 2303–2320.
- [17] G.S. Smith, P.B. Isaacs, The crystal structure of quartz-like GeO₂, *Acta Cryst* 17 (1964) 842–846, <https://doi.org/10.1107/S0365110X64002262>.
- [18] A.T. Dinsdale, A. Akhmetova, A.V. Khvan, N. Aristova, A critical assessment of thermodynamic and phase diagram data for the Ge–O system, *J. Phase Equilib. Diffus.* 36 (3) (2015) 254–261, <https://doi.org/10.1007/s11669-015-0379-1>.
- [19] T. Yamanaka, K. Sugiyama, K. Ogata, Kinetic study of the GeO₂ transition under high pressures using synchrotron X-radiation, *J. Appl. Cryst.* 25 (1) (1992) 11–15, <https://doi.org/10.1107/S0021889891008750>.
- [20] E.E. Nikishina, E.N. Lebedeva, A.V. Piletsky, D.V. Drobot, Hydroxide and oxide of germanium(IV): synthesis and chemical-physical properties, *Fine Chem. Technol.* 10(5) (2015) 19–26 (in Russian). <https://www.finechem-mirea.ru/jour/article/view/254>.
- [21] D.R. Lide (Ed.), *CRC Handbook of Chemistry and Physics*, 89th ed., CRC Press, Boca Raton, 2008.
- [22] Z. Galazka, R. Blukis, A. Fiedler, S. Bin Anooz, J. Zhang, M. Albrecht, T. Remmele, T. Schulz, D. Klimm, M. Pietsch, A. Kwasniewski, A. Dittmar, S. Ganschow, U. Juda, K. Stolze, M. Suendermann, T. Schroeder, M. Bickermann, Bulk single crystals and physical properties of rutile GeO₂ for high-power electronics and deep-ultraviolet optoelectronics, *Phys. Status Solidi B* 2400326 (2024), <https://doi.org/10.1002/pssb.202400326>.
- [23] J. Haines, O. Cambon, E. Philippot, L. Chapon, S. Hull, A neutron diffraction study of the thermal stability of the α -quartz-type structure in germanium dioxide, *J. Solid State Chem.* 166 (2) (2002) 434–441, <https://doi.org/10.1006/jssc.2002.9625>.
- [24] M. Sahnoun, C. Daul, R. Khenata, H. Baltache, Optical properties of germanium dioxide in the rutile structure, *Eur. Phys. J. B* 45 (2005) 455–458, <https://doi.org/10.1140/epjb/e2005-00219-y>.
- [25] M. Labeled, H.J. Jeon, J.H. Park, S.J. Pearton, Y.S. Rim, Rutile germanium dioxide: an emerging ultrawide bandgap semiconductor for power device applications – A review, *Mater. Today* 83 (2025) 513–537, <https://doi.org/10.1016/j.mattod.2025.01.012>.
- [26] S. Chae, L.A. Pressley, H. Paik, D. Werder, B.H. Goodge, L.F. Kourkoutis, R. Hovden, T.M. McQueen, E. Kioupakis, J.T. Heron, Germanium dioxide: A new rutile substrate for epitaxial film growth, *J. Vac. Sci. Technol.* 40 (5) (2022) 050401, <https://doi.org/10.1116/6.0002011>.
- [27] R. Wang, Q. Li, W. Li, P. Jiang, R. Cong, T. Yang, d¹⁰ or d⁰? Theoretical and experimental comparison between rutile GeO₂ and TiO₂ for photocatalytic water splitting, *Chem. Commun.* 57 (4) (2021) 536–539, <https://doi.org/10.1039/D0CC06883G>.
- [28] J.A.S. Laranjeira, S.A. Azevedo, G.S.L. Fabris, A.R. Albuquerque, M.M. Ferrer, J. R. Sambrano, Surface-dependent properties and morphological transformations of rutile GeO₂ nanoparticles, *Appl. Surf. Sci.* 609 (2023) 155321, <https://doi.org/10.1016/j.apsusc.2022.155321>.
- [29] M. Binnewies, R. Glaum, M. Schmidt, P. Schmidt, *Chemical Vapor Transport Reactions*, de Gruyter, Berlin/Boston, 2012.
- [30] G. Schmidt, R. Gruhn, Neue Beobachtungen zum chemischen Transport von GeO₂. VI. Temperaturabhängigkeit mit dem Transportmittel Chlor, *Z. anorg. allg. Chem.* 528(9) (1985) 69–90 (in German). <https://doi.org/10.1002/zaac.19855280908>.
- [31] A.S.T. Rybakov, L.L. Meyer, H. Kuper, M. Chalupa, V. Becker, J.A. Becker, In situ microscopy of composite Germanium/Germanium oxide microneedle growth, *J. Cryst. Growth* 608 (2023) 127133, <https://doi.org/10.1016/j.jcrysgro.2023.127133>.
- [32] C. Dobbe, R. Grotjahn, T. Gebensleben, L.D. Alpei, V. Becker, J.A. Becker, Reactive wetting controlled by very small vertical temperature gradients in a chemical transport mini reactor, *Z. Phys. Chem.* 232 (1) (2018) 79–103, <https://doi.org/10.1515/zpch-2017-0963>.
- [33] MTT Corporation, Important info about corning glass. <https://mtxtl.com/blogs/ceramic-glass-substrate-a-z-info-base/important-info-about-corning-glass>, 2024 (accessed 13 June 2025).
- [34] V.A. Volodin, D.V. Marin, V.A. Sachkov, E.B. Gorokhov, H. Rinnert, M. Vergnat, Applying an improved phonon confinement model to the analysis of Raman spectra of germanium nanocrystals, *J. Exp. Theor. Phys.* 118 (2014) 65–71, <https://doi.org/10.1134/S1063776114010208>.
- [35] C. Fasolato, I. Zardo, M. De Luca, Addressing Crystal Structure in Semiconductor Nanowires by Polarized Raman Spectroscopy, in: N. Fukata, R. Rurali (Eds.), *Fundamental Properties of Semiconductor Nanowires*, Springer, Singapore, 2021, pp. 307–348, https://doi.org/10.1007/978-981-15-9050-4_7.
- [36] M. Kadlecíková, L. Vančo, J. Breza, M. Mikolášek, K. Hušková, K. Fröhlich, P. Procel, M. Zeman, O. Isabella, Raman spectroscopy of silicon with nanostructured surface, *Optik* 257 (2022) 168869, <https://doi.org/10.1016/j.jlleo.2022.168869>.
- [37] S. Weber, *KrystalShaper – Crystal Morphology Editor / Viewer v1.5.0* [software], JCrystalSoft, October 1, 2018. <http://www.jcrystal.com/products/krystalshaper/>.
- [38] C. Boudias, D. Monceau, *CaRIne crystallography v3.1* [software], *CaRIne Crystallography* (1998).
- [39] T. Yamanaka, R. Kurashima, J. Mimaki, X-ray diffraction study of bond character of rutile-type SiO₂, GeO₂ and SnO₂, *Z. Kristallogr. - Cryst. Mater.* 215 (7) (2000) 424–428, <https://doi.org/10.1524/zkri.2000.215.7.424>.
- [40] W.-J. Li, E.-W. Shi, Z.-W. Yin, Growth habit of rutile and α -Al₂O₃ crystals, *J. Cryst. Growth* 208 (1–4) (2000) 546–554, [https://doi.org/10.1016/S0022-0248\(99\)00419-4](https://doi.org/10.1016/S0022-0248(99)00419-4).
- [41] D.A.H. Hanaor, W. Xu, M. Ferry, C.C. Sorrell, Abnormal grain growth of rutile TiO₂ induced by ZrSiO₄, *J. Cryst. Growth* 359 (2012) 83–91, <https://doi.org/10.1016/j.jcrysgro.2012.08.015>.
- [42] Max Planck Institute for Chemical Physics of Solids (2011).
- [43] M. Binnewies, E. Milke, *Thermochemical Data of Elements and Compounds*, second ed., Wiley-VCH, Weinheim, 2002.
- [44] I. Barin, *Thermochemical Data of Pure Substances*, third ed., VCH, Weinheim, 1995.